

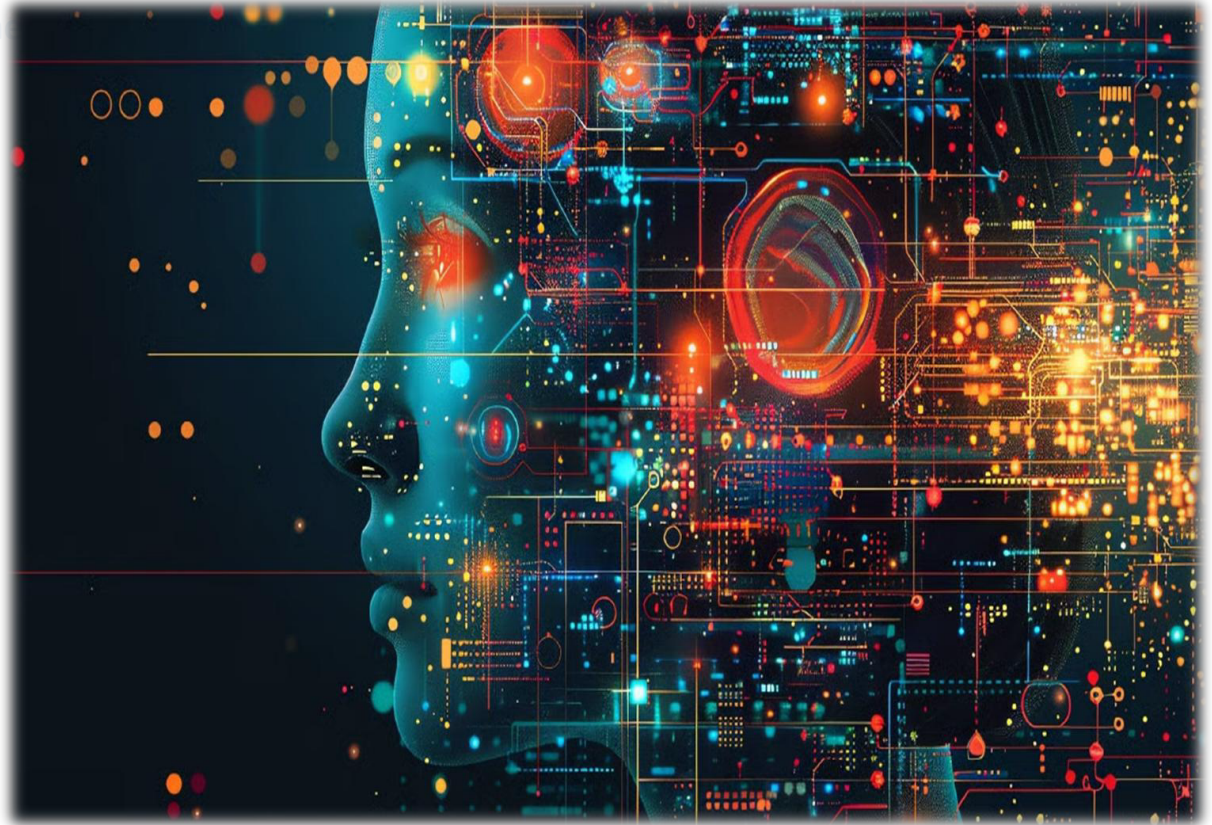
Linear Algebra – The Backbone of AI/ML

Linear Algebra – The Backbone

TOPIC-

Cayley Hamilton Theorem

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C NTENT

- ① C-H Theorem
- ② Uses of C-H Theorem
- ③ Problem based on C-H Theorem

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \rightarrow A^2 = A \cdot A$$

$$\rightarrow A^3 = A^2 \cdot A$$

$$\rightarrow A^{1000} = ?$$

Cayley-Hamilton Theorem

Stnt:- Every square matrix satisfies its own characteristic equation.

Uses of Cayley-Hamilton Theorem

To calculate —

- (i) The positive integral powers of A and
- (ii) The inverse of a non-singular matrix of A .

Example

(1) show that the matrix $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ satisfies its own characteristic equation.

Sol:- Let $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

The char. equation of the given matrix is $|A - \lambda I| = 0$

$$\Rightarrow \lambda^2 - s_1 \lambda + s_2 = 0$$

Where $S_1 = \text{Sum of the main dia. elements}$

$$= 1+1=2$$

$$S_2 = \det |A|$$

$$= |A| = \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = 1+4 = 5$$

$\therefore$ The characteristic eq.

$$\lambda^2 - 2\lambda + 5 = 0$$

To prove: $A^2 - 2A + 5I = 0$ — (1)

$$A^2 = A A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} -3 & -4 \\ 4 & -3 \end{pmatrix}$$

$$2A = 2 \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}$$

So from — (1)

$$A^2 - 2A + 5I$$

$$= \begin{pmatrix} -3 & -4 \\ 4 & -3 \end{pmatrix} - \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \text{proved}$$

$\therefore$ Therefore Mat. A satisfies its ch. eqn.

② Verify the C-H Theo. for $A = \begin{pmatrix} 3 & 5 \\ -2 & -4 \end{pmatrix}$

Solⁿ ∴

Now, $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 5 \\ -2 & -4-\lambda \end{vmatrix} = 0$$

$$\lambda^2 + \lambda - 2 = 0$$

$A^2 + A - 2I = 0$ to verify
The Cayley-Hamilton Theorem.
Therefore,

$$A^2 + A - 2I = \begin{pmatrix} 3 & 5 \\ -2 & -4 \end{pmatrix}^2 + \begin{pmatrix} 3 & 5 \\ -2 & -4 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & -5 \\ 2 & 6 \end{pmatrix} + \begin{pmatrix} 3 & 5 \\ -2 & -4 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \text{proved,}$$

③ H/W

Verify the Cayley-Hamilton theorem
for $A = \begin{pmatrix} 3 & 4 \\ -5 & -5 \end{pmatrix}$

④ Use The C-H Theorem to calculate the inverse of $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$.

Solⁿ: $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix}$
 $= (1-\lambda)(3-\lambda) - (4)(2)$
 $= \lambda^2 - 4\lambda - 5$

By the C-H Theorem, A satisfies:

$$A^2 - 4A - 5I = 0$$
$$\bar{A}^1 (A^2 - 4A - 5I) = 0$$
$$\Rightarrow A - 4I - 5\bar{A}^1 = 0$$
$$\Rightarrow 5\bar{A}^1 = A - 4I$$
$$\Rightarrow \bar{A}^1 = \frac{1}{5} \{ A - 4I \}$$
$$\Rightarrow \bar{A}^1 = \frac{1}{5} \left\{ \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \right\}$$

$$\bar{A}^1 = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ 2 & -1 \end{pmatrix}$$

5) Let $A = \begin{pmatrix} 3 & -2 & -1 \\ 2 & 3 & 4 \\ -2 & 0 & 5 \end{pmatrix}$ Use the C-H Theorem to calculate A^{-1} and A^4 .

Solⁿ:- $|A - \lambda I| = \begin{vmatrix} 3-\lambda & -2 & -1 \\ 2 & 3-\lambda & 4 \\ -2 & 0 & 5-\lambda \end{vmatrix}$ and

$$|A - \lambda I| = (3-\lambda) \begin{vmatrix} 3-\lambda & 4 \\ 0 & 5-\lambda \end{vmatrix} - (-2) \begin{vmatrix} 2 & 4 \\ -2 & 5-\lambda \end{vmatrix} - 1 \begin{vmatrix} 2 & 3-\lambda \\ -2 & 0 \end{vmatrix}$$

$$= (3-\lambda) [(3-\lambda)(5-\lambda) - 0] + 2[10 - 2\lambda + 8] - 1[0 + 2(3-\lambda)]$$

$$= 45 - 9\lambda - 30\lambda + 6\lambda^2 + 5\lambda^2 - \lambda^3 + 36 - 4\lambda - 6 + 2\lambda$$

$$= -\lambda^3 + 11\lambda^2 - 41\lambda + 75$$

By C-H Theorem, A satisfies:-

$$-A^3 + 11A^2 - 41A + 75I = 0$$

$$\bar{A}'(-A^3 + 11A^2 - 41A + 75I) = 0$$

$$-A^2 + 11A - 41I + 75\bar{A}' = 0$$

$$75\bar{A}' = A^2 - 11A + 41I$$

$$\bar{A}' = \frac{1}{75} \{ A^2 - 11A + 41I \}$$

$$\bar{A}' = \frac{1}{75} \left\{ \begin{pmatrix} 3 & -2 & -1 \\ 2 & 3 & 4 \\ -2 & 0 & 5 \end{pmatrix}^2 - 11 \begin{pmatrix} 3 & -2 & -1 \\ 2 & 3 & 4 \\ -2 & 0 & 5 \end{pmatrix} + \begin{pmatrix} 41 & 0 & 0 \\ 0 & 41 & 0 \\ 0 & 0 & 41 \end{pmatrix} \right\}$$

$$\bar{A}' = \frac{1}{75} \begin{pmatrix} 15 & 10 & -5 \\ -18 & 13 & -14 \\ 6 & 4 & 13 \end{pmatrix}$$

Also,

$$A(-A^3 + 11A^2 - 41A + 75I) = 0$$

$$\Rightarrow -A^4 + 11A^3 - 41A^2 + 75A = 0$$

Thus,

$$A^4 = 11A^3 - 41A^2 + 35A$$

$$A^4 = \begin{pmatrix} 319 & -550 & -1485 \\ -418 & 77 & 1826 \\ -1034 & 484 & 1837 \end{pmatrix} + \begin{pmatrix} -287 & 492 & 656 \\ -164 & -205 & -1236 \\ 656 & -164 & -1107 \end{pmatrix} + \begin{pmatrix} 225 & -150 & -75 \\ 150 & 225 & 300 \\ -150 & 0 & 375 \end{pmatrix}$$

$$A^4 = \begin{pmatrix} 257 & -208 & -909 \\ -432 & 97 & 896 \\ -528 & 320 & 1105 \end{pmatrix}$$

⑥ Let $A = \begin{pmatrix} 3 & 1 \\ 5 & -2 \end{pmatrix}$. Use the C-H Theorem to find the constants a and b such that $A^3 = aA + bI$

Solⁿ:- $A - \lambda = \begin{pmatrix} 3-\lambda & 1 \\ 5 & -2-\lambda \end{pmatrix}$ and

$$(A - \lambda) = (3 - \lambda)(-2 - \lambda) - 5(1).$$

$$= d^2 - \lambda - 11$$

By the C-H theorem, A satisfies:

$$A^2 - A - 11I = 0 \text{ therefore,}$$

$$A(A^2 - A - 11I) = 0$$

$$A^3 - A^2 - 11A = 0$$

$$A^3 = A^2 + 11A$$

$$A^3 = A + 11I + 11A$$

$$A^3 = 12A + 11I$$

Therefore $a = 12, b = 11$

⑦ Let $A = \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$. Use the C-H theorem to formulate A^{n+2} .

⑧ How to calculate the power of any matrix using C-H theorem

Q:- $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$
 Find. $A^9 = ?$

Assume, $A^9 = C_1 A + C_2 I$ — (i)

replace by λ constant
 $I = I$ Identity Matrix
 $\lambda^9 = C_1 \lambda + C_2$ — (ii)

Now, by solving characteristic equation we get, $\lambda = 1$ or 2

For $\lambda = 1$

$$(1)^9 = C_1 \cdot (1) + C_2$$

$$C_1 + C_2 = 1 \text{ — (iii)}$$

For $\lambda = 2$

$$(2)^9 = C_1 \cdot (2) + C_2$$

$$2C_1 + C_2 = 512 \text{ — (iv)}$$

Solving — (iii) & (iv)

$$C_1 = 511$$

$$C_2 = -510$$

Putting the value of c_1 & c_2 in - (1)

$$A^9 = 511A - 510I$$

Q. $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$

Find the characteristic equation of A and hence find A^{-1} .

Solⁿ:-

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda) [(1-\lambda)(2-\lambda)] + \lambda - 1 = 0$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

$$\boxed{A^3 - 5A^2 + 7A - 3I = 0}$$

$\downarrow$
Ch. equation

Now, multiply $\bar{A}^{-1}$ on both sides,

$$\bar{A}^{-1} \cdot A^3 - 5\bar{A}^{-1} \cdot A^2 + 7\bar{A}^{-1} \cdot A - 3\bar{A}^{-1} \cdot I = \bar{A}^{-1} \cdot 0$$

$$A^2 - 5A + 7I - 3\bar{A}^{-1} = 0$$

$$A^2 - 5A + 7I = 3\bar{A}^{-1}$$

$$\bar{A}^{-1} = \frac{1}{3} \{ A^2 - 5A + 7I \}$$

$$= \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & 1 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

HW-1

Consider a 3×3 matrix

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Use C-H Theorem and verified?

HW-2

Let F be a 2×2 Matrix and given
by $F = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and Find (F^3)
Using C-H Theorem

Consider a 3×3 Matrix $(D) = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$

- Find the characteristic polynomial of Matrix (D) .
- Applying C-H Theorem to show that $D^3 - 6D^2 + 12D - 8I = 0$

H/w-4

Let Matrix $J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

- Determine the ch. polynomial of Matrix (J)

- Use the C-H Theorem to verify that $J^2 - 4J + 4I = 0$

H/W-5

Consider a 3×3 matrix (K) :

$$K = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix}$$

- Find the characteristic polyno. of matrix (K) -
- Apply the C-H Theorem to show that $K^3 - 15K^2 + 75K - 125I = 0$

Define $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

verify C-H Theorem?